

Game

Bibi and Jojo are playing a game called "Upper or Lower". In this game, Bibi will give Jojo array of characters then Bibi will say the query and Jojo must change the case of the character at the index of the query (index is 0-based). If a character is in it's lower case, change it to the upper case and If a character is in it's upper case, change it to the lower case

Format Input

First line of input consists of integer N and M. N is the length of the array of characters and M is the total number of query. The next line consist N-length array of character. Last, the next M line consist the query. It's guaranteed that the N-length array of characters does not contain space.

Format Output

Output the resulting string after all the queries.

Constraints

$0 < N < 201$

$0 < M < 101$

Sample Input 1	Sample Output 1
5 3 Hello 0 1 3	hElLo

Sample Input 2	Sample Output 2
10 7 BIBIJOJOLI 0 3 4 8 2 3 9	bIbIjOjOli

Sample Input 3	Sample Output 2
5 3 Hello 0 0 1	HEllo